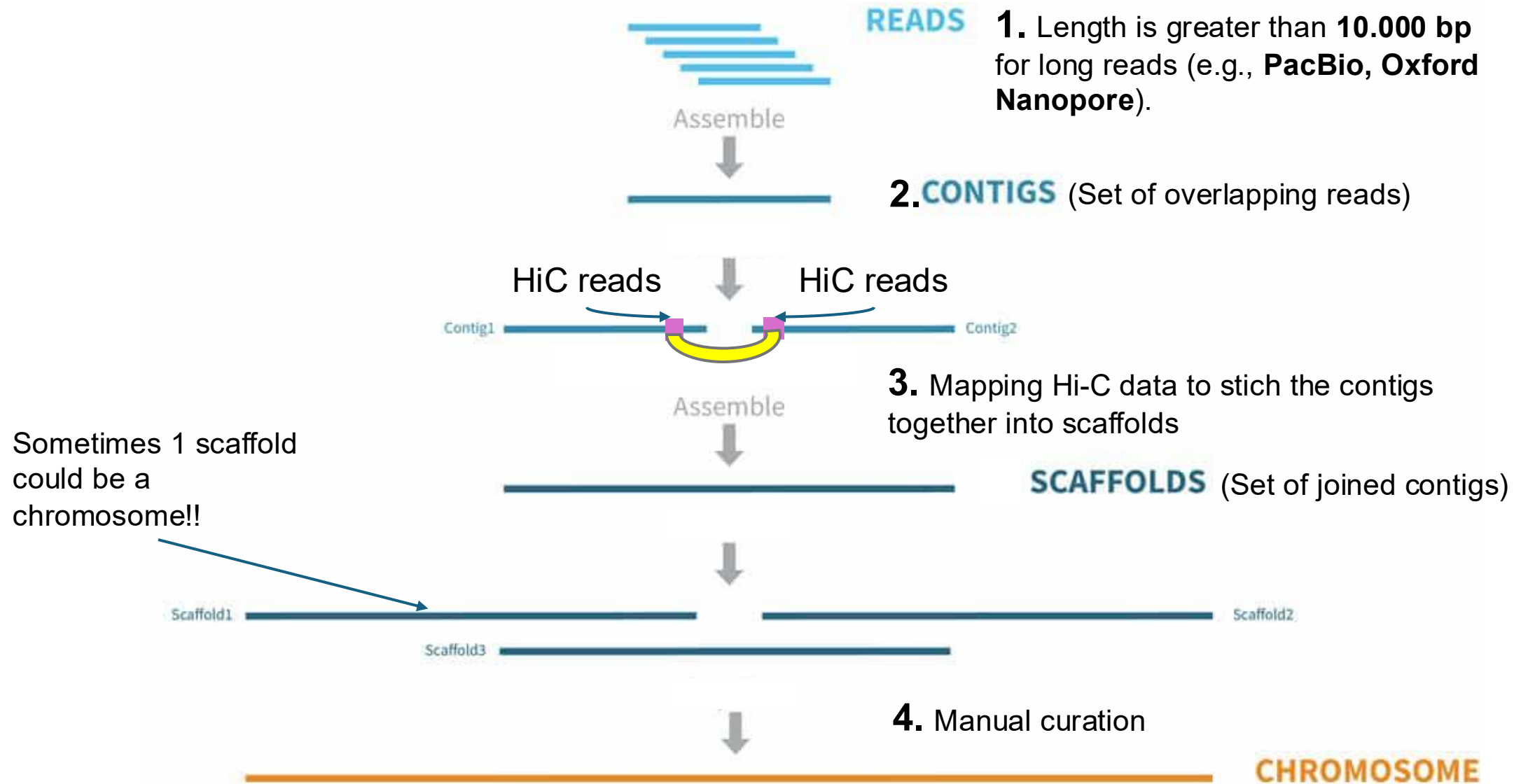


Genome assembly
Biodiversity genomics
course
Mendoza-Argentina
2025

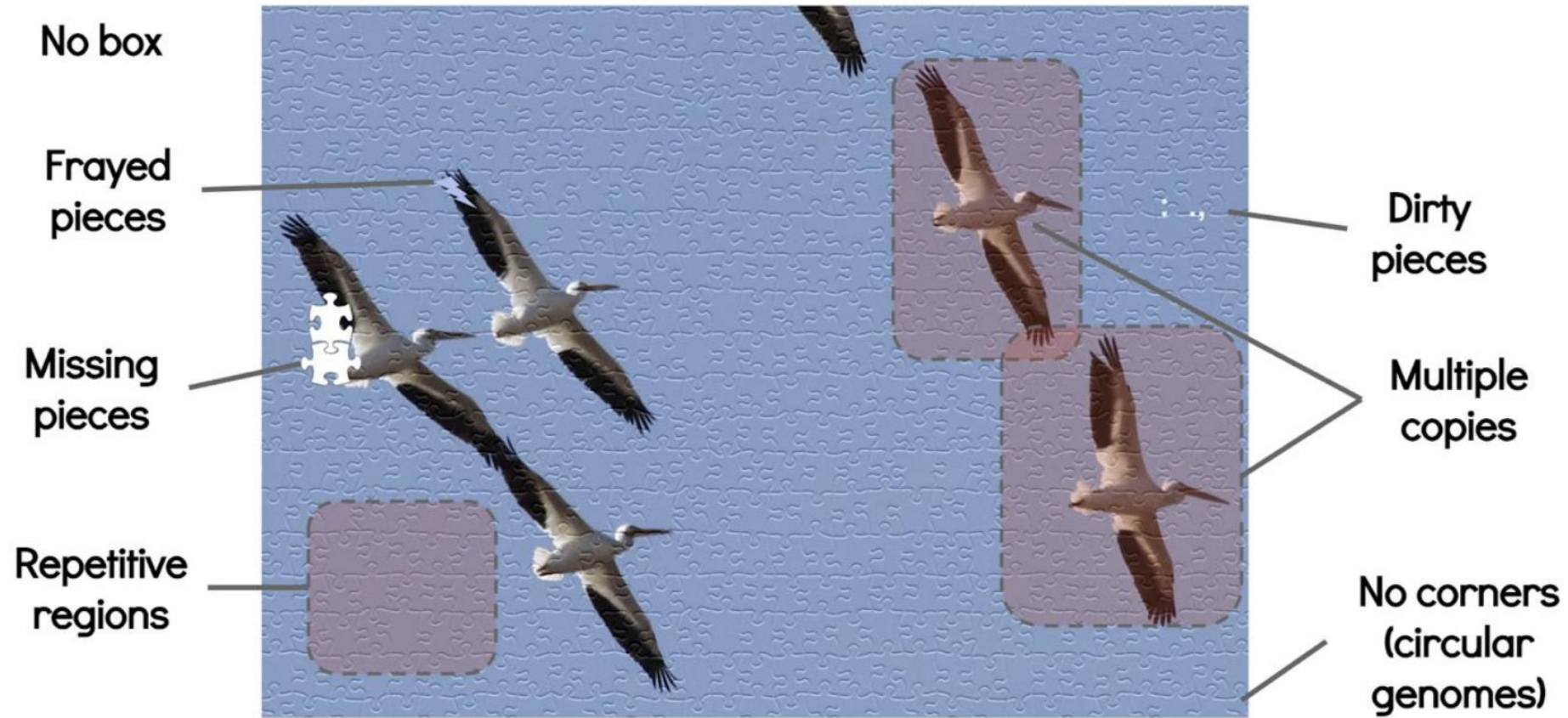


General steps genome assembly



Why is important to use long reads?

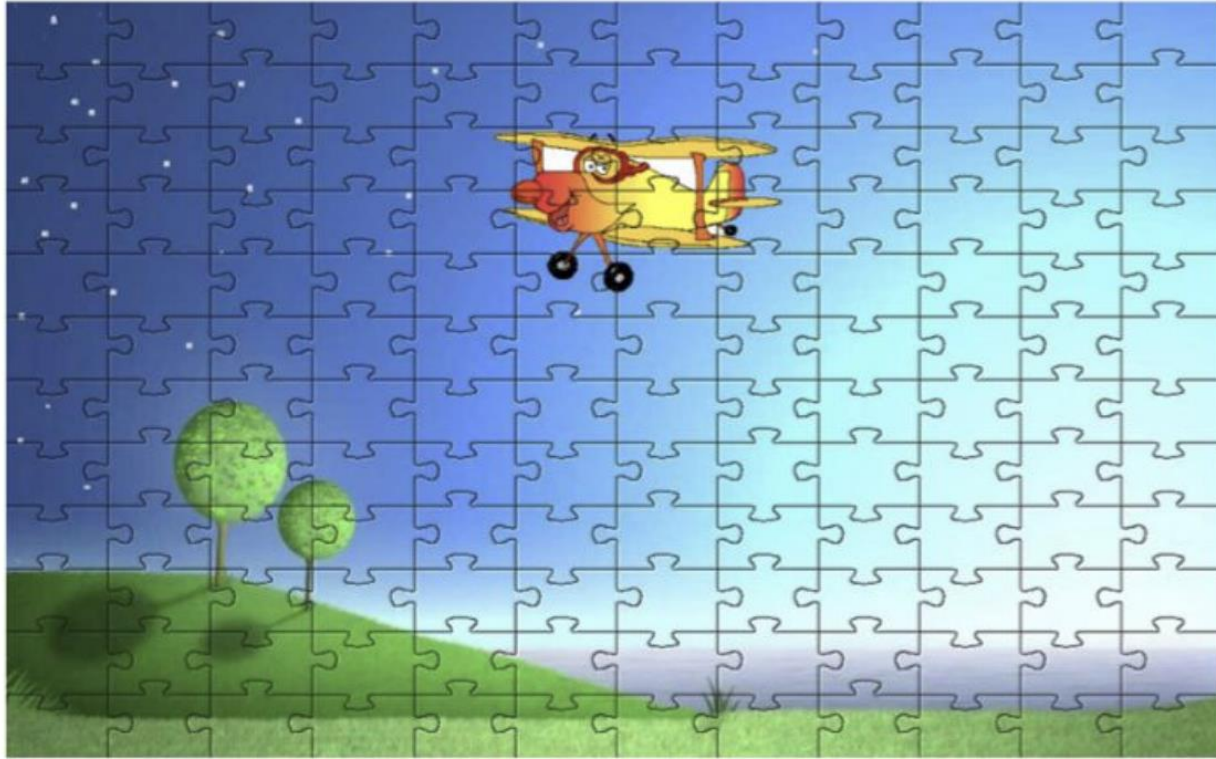
What makes a jigsaw puzzle hard?



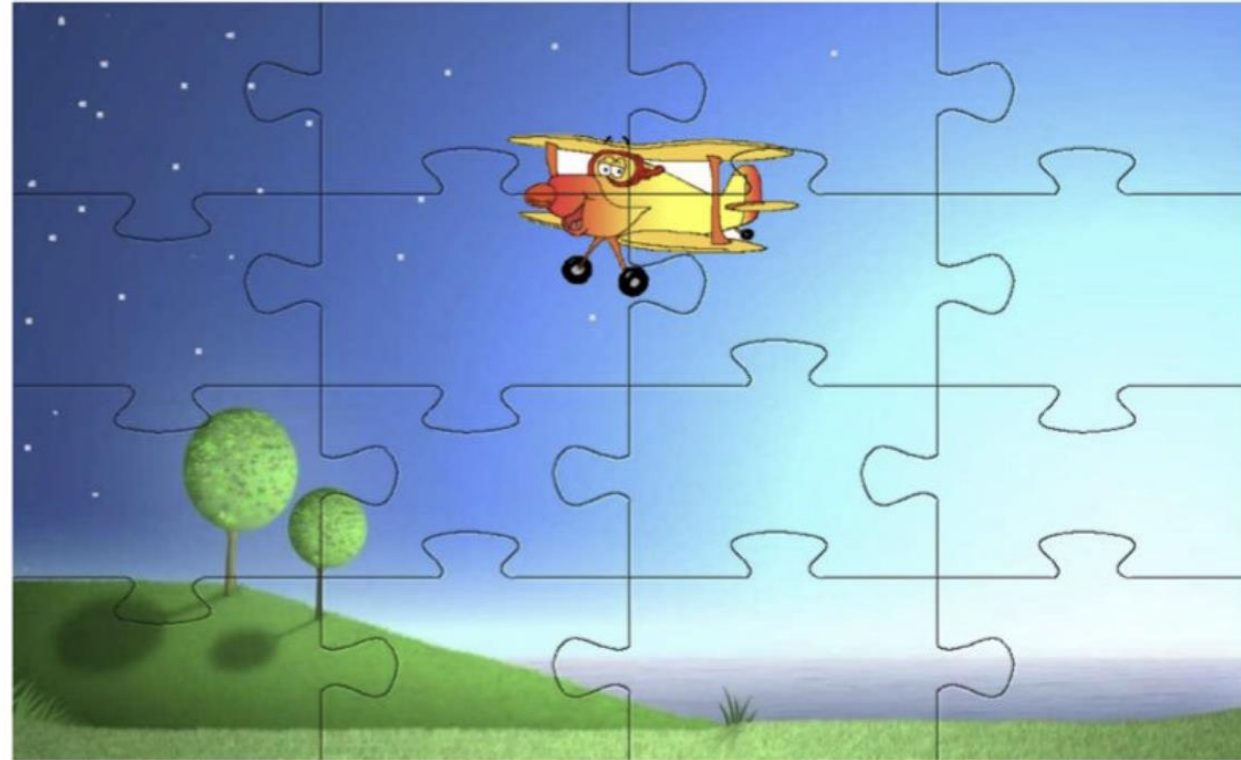
- What helps? Larger pieces (read length); fewer dirty or frayed pieces (errors in reads). fewer repeats and copies...

Why is important to use long reads?

Genome assembly with short reads



PacBio, Oxford Nanopore
Genome assembly with long reads



Step 2. Assembling the reads

Aligned reads

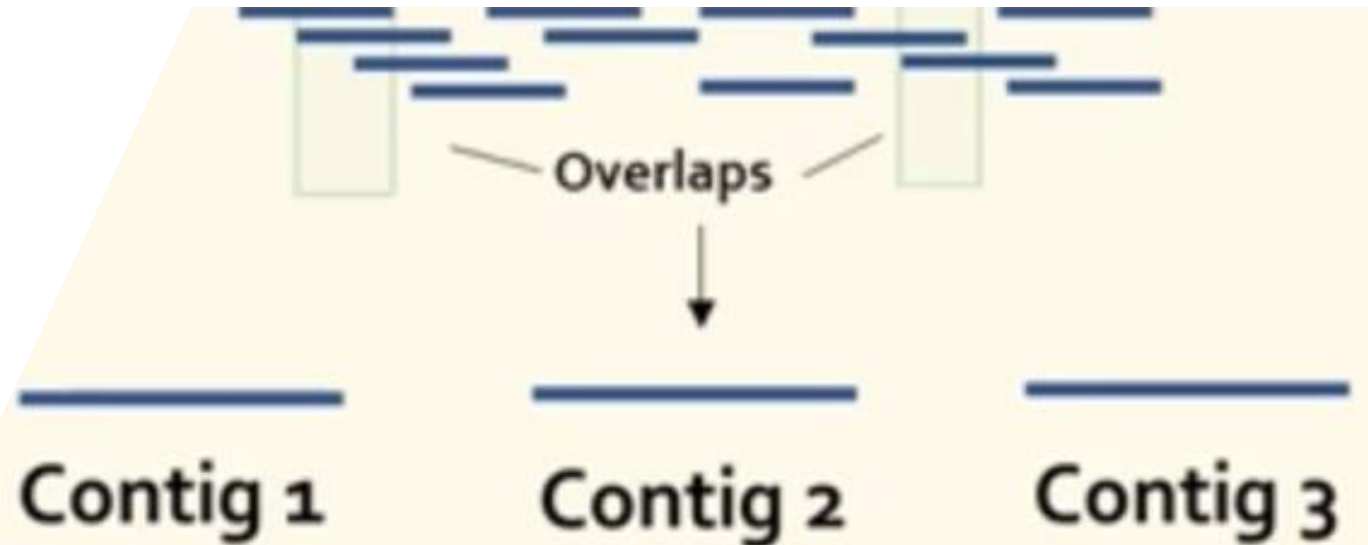
```
TGAAGTCCTACAGTCATAGTC
  AAGTCCTACAGTCATAGTCGA
    GTCCTACAGTCATAGTCGATA
      CCTACAGTCATAGTCGATATT
        TACAGTCATAGTCGATATTT
```

Consensus contig TGAAGTCCTACAGTCATAGTCGATATTT

Although the contigs are longer than reads and do not have gaps, they are typically not large enough to cover entire chromosomes because some parts of chromosomes are very repetitive or otherwise hard to sequence trough.

Software:

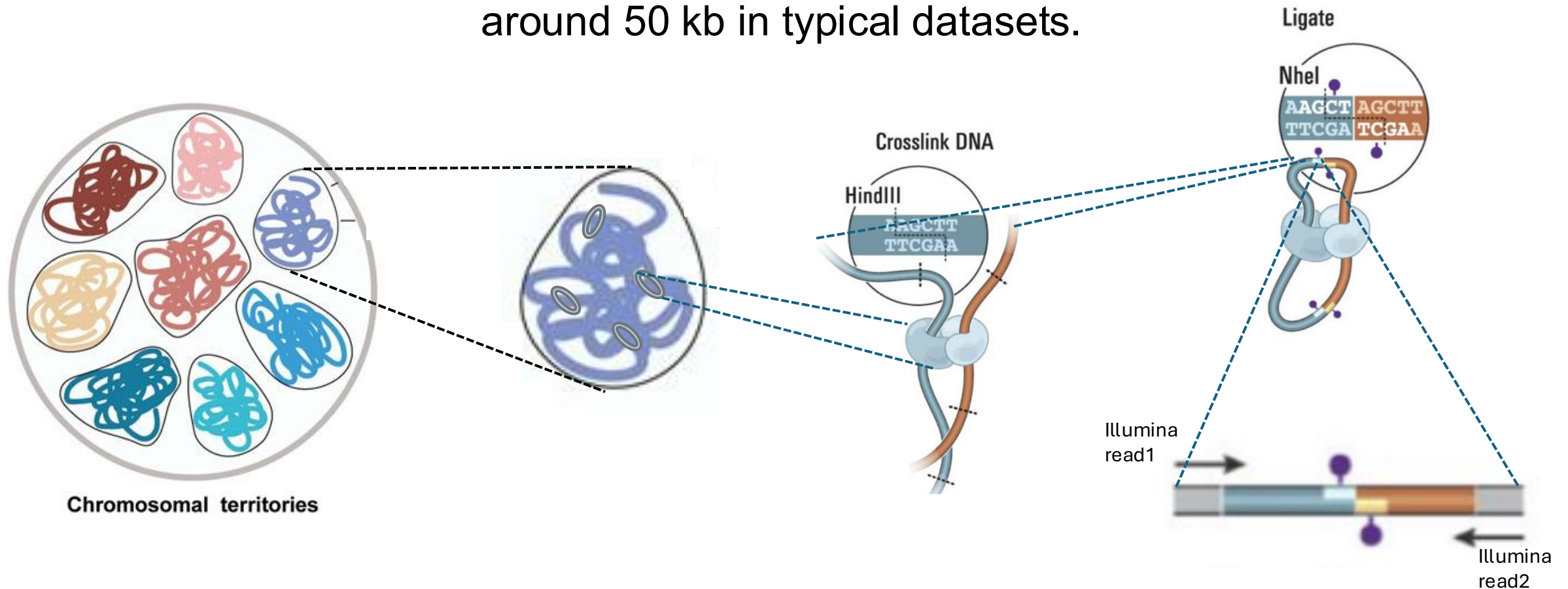
HiFi
Canu
Trinity
SPAdes
Flye



Step 3. Scaffolding (Assembling the contigs)

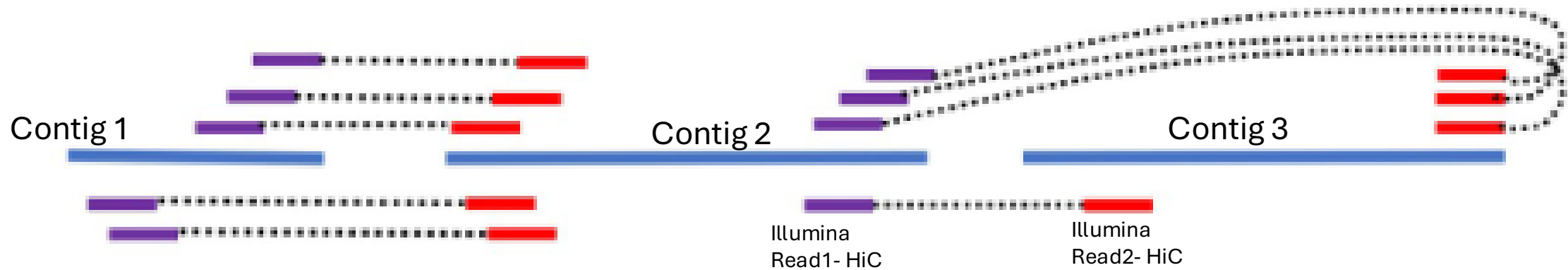
Hi-C data

Aim: Join two nearby parts of a chromosome together and then sequence across the join to get reads from regions of the same chromosome, often tens to hundreds of kilobases apart—for example, around 50 kb in typical datasets.



Step 3. Scaffolding (Assembling the contigs)

1. Mapping HiC data to Contigs:



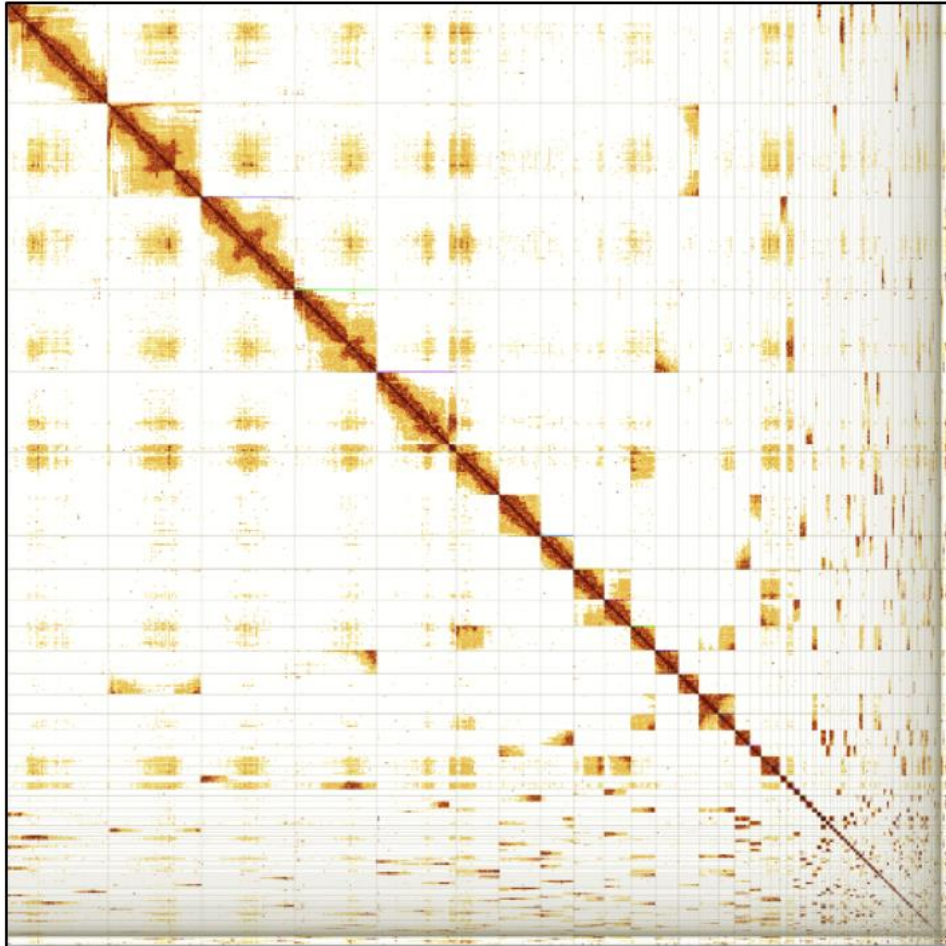
2. Stitching contigs together to form scaffolds:



Step 4. Genome curation

n = 230

N50 = 33.1Mb

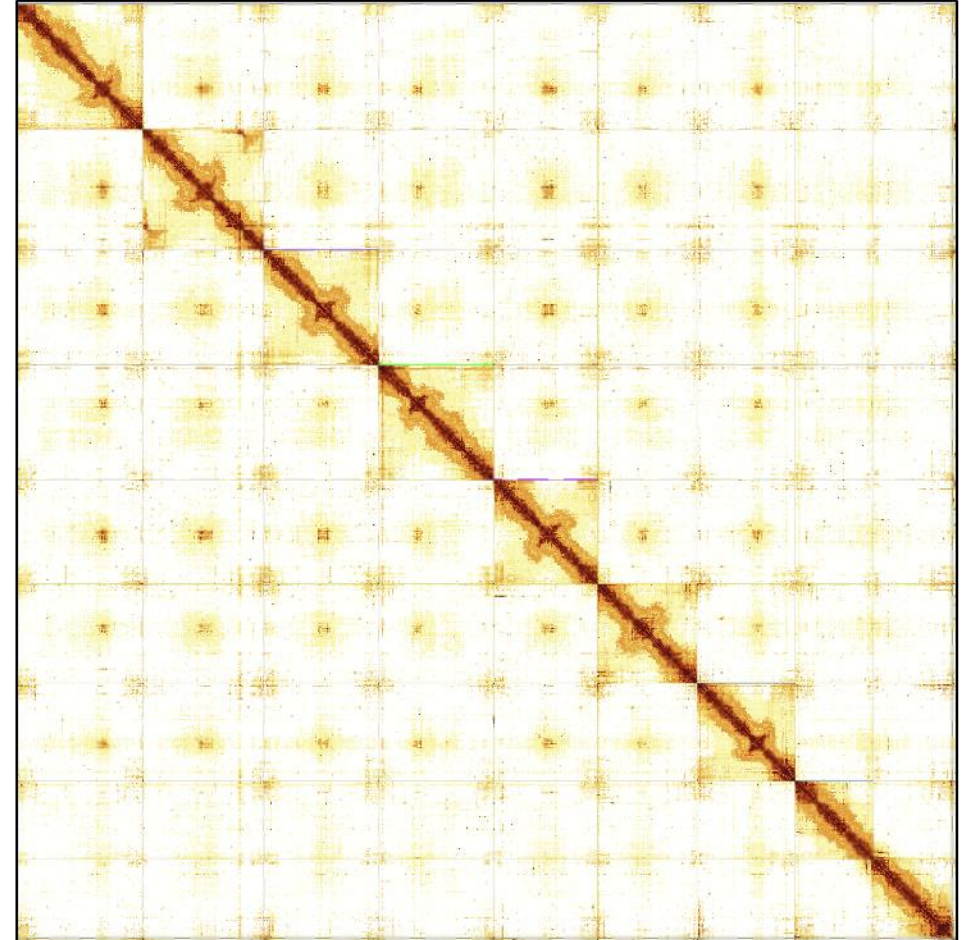


Pre curation assembly

n = 62

N50 = 87.1Mb

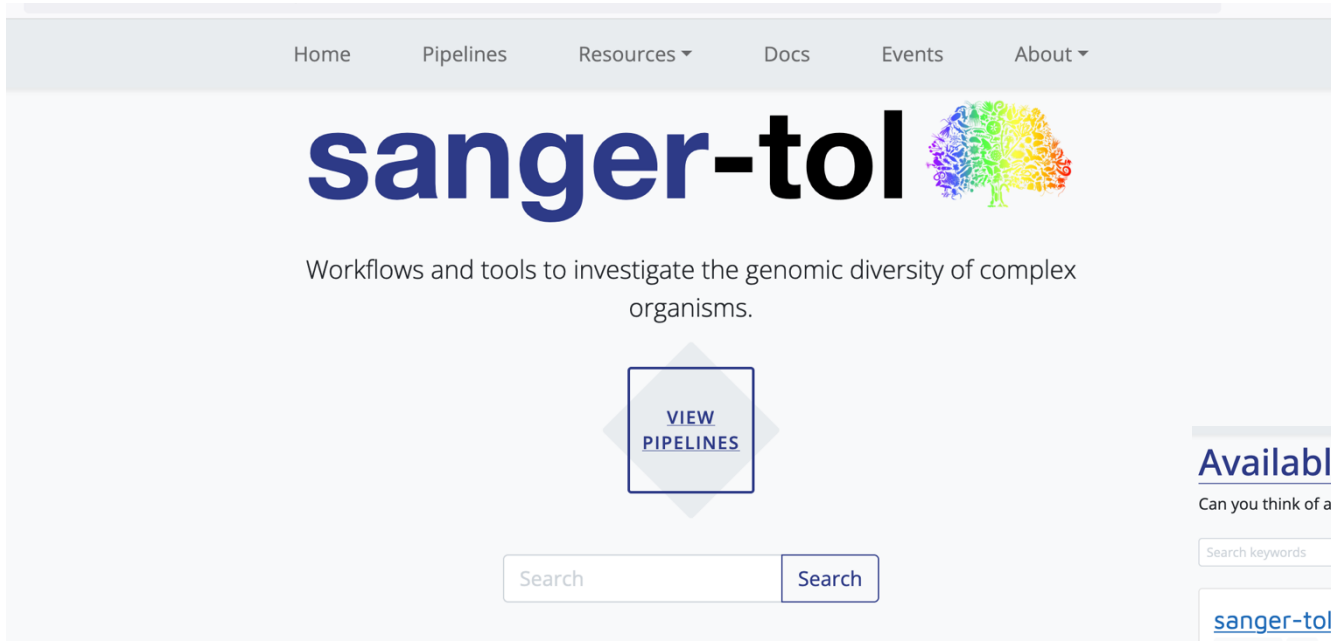
99.85% of genome in 9 Chromosomes



Post curation

Pipelines to generate a genome

<https://pipelines.tol.sanger.ac.uk/>



Sanger-Tol recipe is working
across the Tree of life
4000 species

Chromosome level genomes

- 25x Pacbio HiFi
- 100x Hi-C (Arima/Qiagen)

Available Pipelines

Can you think of another pipeline that would fit in well? [Let us know!](#)

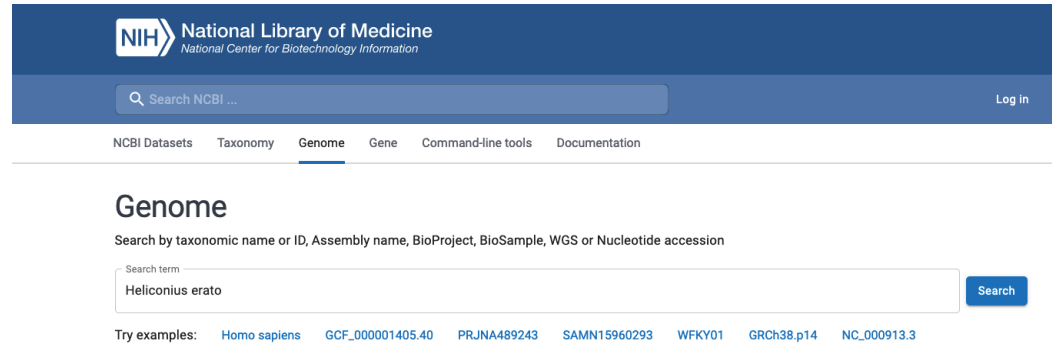
Search keywords: Filter: Released **10** Under development **4** Sort: Last Release Alphabetical Stars Display: ☐

sanger-tol/curationpretext 3 genomics hic A Nextflow DSL2 pipeline for pretext generation in curation Version 1.0.0 Published 2 days ago	sanger-tol/treeval curation genome-alignment genome-assembly genomics quality-control synteny Pipelines for the production of Treeval data Version 1.1.1 Published 2 weeks ago
sanger-tol/ensemblgenedownload 1 download gene-annotation genomics indexing Nextflow pipeline to download gene annotations from Ensembl onto the Tree of Life directory structure Version 2.0.0 Published 2 months ago	sanger-tol/ensemblrepeatdownload download formatting genomics Nextflow pipeline to download repeat annotations from Ensembl onto the Tree of Life directory structure Version 2.0.0 Published 2 months ago
sanger-tol/insdcdownload 1 download genomics indexing Nextflow pipeline to download assemblies from INSDC and add them to the Tree of Life directories Version 2.0.0 Published 2 months ago	sanger-tol/variantcalling alignment genomics variant-calling This Nextflow DSL2 pipeline calls variants on long read alignment. It is run after sanger-tol/readmap in the Sanger ToL production suite but with options to run on unaligned reads. Version 1.1.3 Published 2 months ago

Where can we find genomes?

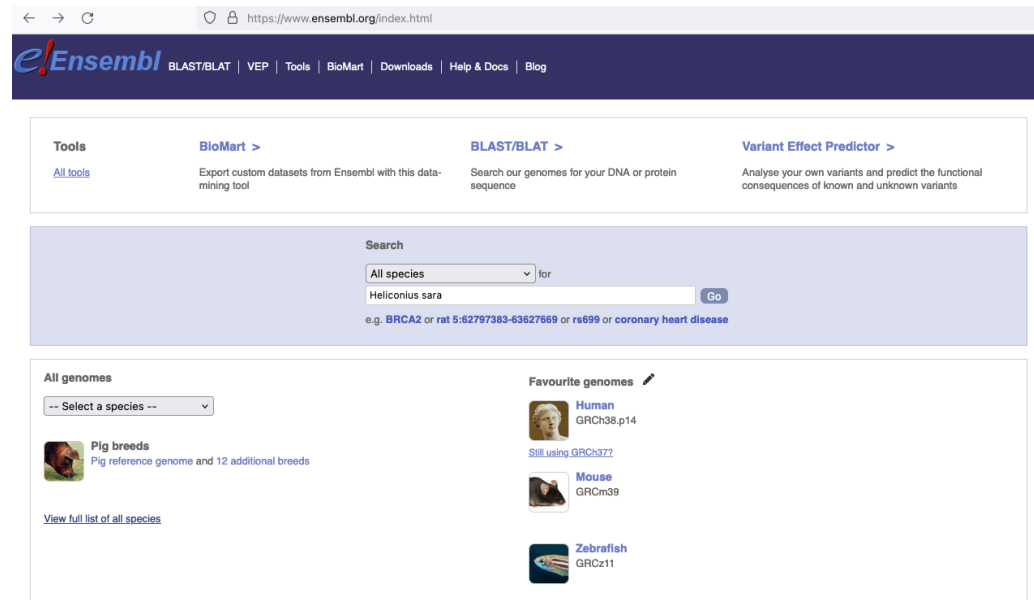
- NCBI (National Center for Biotechnology Information):

<https://www.ncbi.nlm.nih.gov/datasets/genome/>



- ensembl

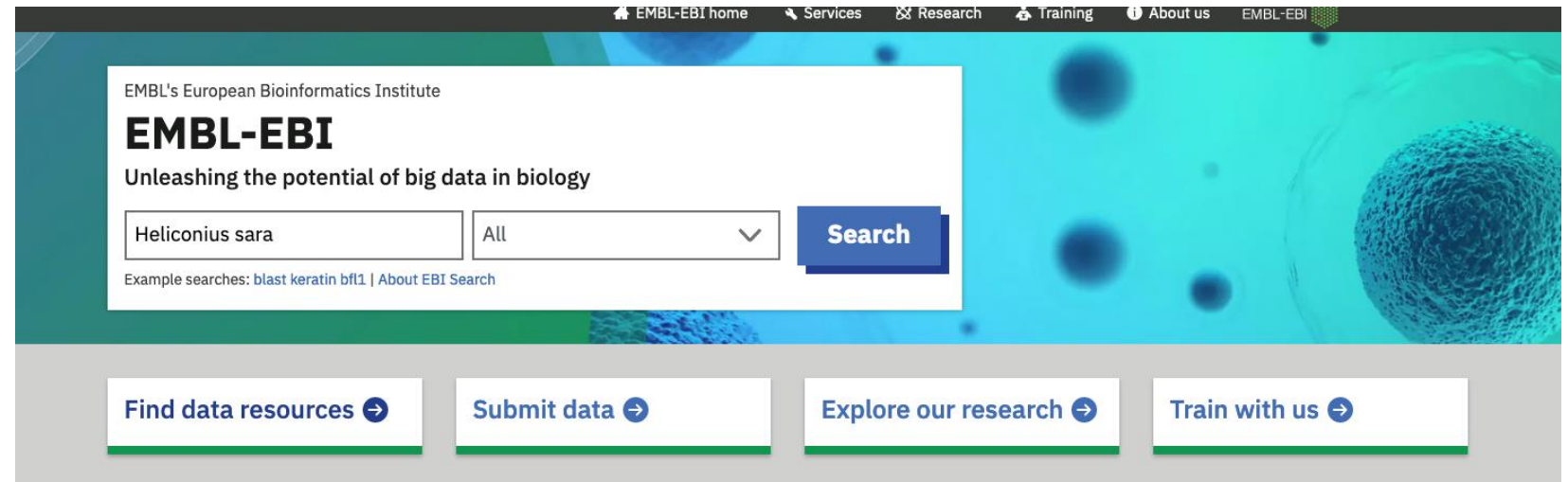
<https://www.ensembl.org/index.html>



Where can we find genomes?

- EMBL-EBI (European Molecular Biology Laboratory - European Bioinformatics Institute):

<https://www.ebi.ac.uk/>



Example

https://www.ncbi.nlm.nih.gov/datasets/genome/

An official website of the United States government [Here's how you know](#)

National Library of Medicine
National Center for Biotechnology Information

Log in

NCBI DatasetsTaxonomyGenomeGeneCommand-line toolsDocumentation

Genome

Search by taxonomic name or ID, Assembly name, BioProject, BioSample, WGS or Nucleotide accession

Search term

Heliconius sara

×

Search

Try examples: [Homo sapiens](#) [GCF_000001405.40](#) [PRJNA489243](#) [SAMN15960293](#) [WFKY01](#) [GRCh38.p14](#) [NC_000913.3](#)

https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_917862395.2/

An official website of the United States government [Here's how you know](#)

National Library of Medicine
National Center for Biotechnology Information

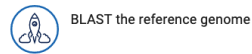
Log in

NCBI DatasetsTaxonomyGenomeGeneCommand-line toolsDocumentation

Genome assembly iHelSar1.2 reference

Download	 datasets	URL	FTP	Actions
Submitted GenBank assembly	GCA_917862395.2			⋮
Taxon	Heliconius sara			
WGS project	CAKJTV02			
Assembly type	haploid			
Submitter	WELLCOME SANGER INSTITUTE			
Date	Apr 6, 2023			

View the [legacy Assembly page](#)



Additional genomes

[Browse all Heliconius sara genomes \(2\)](#)

BioProject

[PRJEB46449](#)

Heliconius sara (Sara longwing) genome assembly iHelSar1

https://www.ncbi.nlm.nih.gov/datasets/genome/?taxon=33443

An official website of the United States government [Here's how you know](#)

National Library of Medicine
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Genome

Download a genome data package including genome, transcript and protein sequence, annotation and a data report

Selected taxa

[Heliconius sara](#)  Enter one or more taxonomic names

Filters

Download Select columns 2 Genomes 1 selected Rows per page 20 1-2 of 2

☐	Assembly	GenBank	RefSeq	Scientific name	Modifier	Annotation	Action
☒	iHelSar1.2		GCA_917862395.2	Heliconius sara			⋮
☐	iHelSar1.1 alternate haplotype...		GCA_911392485.1	Heliconius sara			⋮

https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/917/862/395/GCA_917862395.2_iHelSar1.2/

Index of /genomes/all/GCA/917/862/395/GCA_917862395.2_iHelSar1.2

Name	Last modified	Size
Parent Directory		-
GCA_917862395.2_iHelSar1.2_assembly_report.txt	2023-04-18 17:05	34K
GCA_917862395.2_iHelSar1.2_assembly_stats.txt	2023-04-18 17:05	31K
GCA_917862395.2_iHelSar1.2_feature_count.txt	2024-02-15 01:58	168
GCA_917862395.2_iHelSar1.2_genomic.fna.gz	2023-04-18 17:05	107M
GCA_917862395.2_iHelSar1.2_genomic.gbff.gz	2023-04-18 17:05	138M
GCA_917862395.2_iHelSar1.2_genomic_gaps.txt.gz	2023-04-18 17:05	1.4K
GCA_917862395.2_iHelSar1.2_wgsmaster.gbff.gz	2023-04-18 17:05	1.4K
README.txt	2024-04-11 16:11	54K
annotation_hashes.txt	2024-02-17 10:02	410
assembly_status.txt	2024-07-15 13:53	14
md5checksums.txt	2024-02-17 10:02	625
uncompressed_checksums.txt	2024-06-14 01:21	167

[HHS Vulnerability Disclosure](#)

wget https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/917/862/395/GCA_917862395.2_iHelSar1.2/GCA_917862395.2_iHelSar1.2_genomic.fna.gz

Reference genomes are stored in a fasta format

It is a text-based format for representing nucleotide sequences (DNA or RNA)

```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6) ← Description Line
ACCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCCGGGCTCCGGCCCCGGCCCCGGCTCGGGGCCCCGCGGGGCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCCGCCCAAGTGGCCCCGGGGCTTGATTTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT ← Sequence Lines
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGGTCGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

Description Line: Begins with a > character followed by an identifier and optional description.

Sequence Lines: The actual sequence data, typically represented over multiple lines for readability.

Assembly statistics

	GenBank
Genome size	562.2 Mb
Total ungapped length	562.2 Mb
Number of contigs	203,084
Contig N50	9.9 kb
Contig L50	14,088
GC percent	33.5
Genome coverage	1.9x
Assembly level	Contig

It is the median of the contig lengths.

It is the number of contigs (or scaffolds) that together contain at least 50% of the total assembly length.

Significance

N50: A higher N50 value indicates that the assembly contains longer contigs, suggesting better assembly quality and continuity.

L50: A lower L50 value indicates that fewer contigs are needed to cover 50% of the genome, suggesting higher assembly contiguity.

References

- Jay Ghurye and Mihai Popl. 2019. Modern technologies and algorithms for scaffolding assembled genomes. Plos computational biology
- Edward S. Rice and Richard E. Green. 2018. New Approaches for Genome Assembly and Scaffolding. Annual Review of Animal Biosciences.
- https://ucdavis-bioinformatics-training.github.io/2020-Genome_Assembly_Workshop/kmers/kmers
- <https://www.youtube.com/watch?v=8w-CbyGV-pgv>
- <https://star-protocols.cell.com/protocols/1799>
- <https://training.galaxyproject.org/training-material/topics/sequence-analysis/tutorials/quality-control/tutorial.html#per-sequence-quality-scores>